REQUIREMENT DESCRIPTION WITH SALIENT CHARACTERISTICS

736 SFS Radar Payload EchoGuard Panels or Equal

3D RADAR PAYLOAD ECHOGUARD PANELS OR EQUAL

The current ODIN Tower System is currently in need of an upgrade of EchoGuard radar panels in order to increase and enhance detection capabilities to small unmanned aerial systems, vehicles, personnel among other objects. This EchoGuard radar panels should be packaged in hard body transit cases for deployment operations and must have capabilities for slew to cue integration with current ODIN dual camera system with fix thermal lense for immediate target identification with 120 degrees field of view (360 degrees of coverage total) with Range, Azimuth and Elevation identification. Must include all necessary mounting hardware for ODIN Tower System integration. Provide heightened cyber-security, enable software release updates to ensure system is not vulnerable to attacks. These items must be compatible to work and fit DFT's ODIN Tower system. Vendor must offer a minimum of a one (1) yr warranty/training support and local in-person training support OCONUS within a year of purchase. Vendor must be able to work on current ODIN Sensor Tower/PTZ day/night dual camera with fixed Thermal lens, excluding radar package - TAK Person Worn Kit and mCore Edge Processor suite with laptop and system accessories.

1. Remotely operated radar system
2. Camera integration and sensor feed to an operation center
3. Must have capabilities for slew to cue integration
4. Hard body transit cases
5. Able to update software to reduce cyber security threat
6. ATAK system integration
7. Smart phone or tablet compatibility, Command and control
8. Ability to Geo-Locate
9. Capability for 120 degrees detection to exceed 800 meters
10. Capability for 360 detection to exceed 800 meters
11. Rugged construction
12. Must be able to provide range, azimuth and elevation identification
13. Frequency
    a. K-band 24.45 – 24.65 GHz (USA)
    b. K-band 24.05 – 24.25 GHz (INTL)
14. Field of View- 120° Azimuth x 80° Elevation
15. Track Accuracy - < 1° Azimuth x < 1.5° Elevation
16. Track Update Rate - 10 Hz
17. Max Tracks - Up to 20 Simultaneous Tracks
18. Instrumented Range - 6 km Range

19. Short-Range, Software-Defined, Metamaterials ESA (MESA®) Radar in a Compact Solid-State Format.
20. Must have Unmanned Aerial System detection capabilities.
21. Must have MPU5 radio integration capabilities.
22. Range
    sUAV: > 1 km (Phantom 4)
    1.4 km (Matrice 600)
    Vehicle: > 3.5 km
    Human: > 2.2 km
23. Size -16.3 cm x 20.3 cm x 5.7 cm
    Weight - 1.25 kg
    Power (USA): + 15 to + 28 VDC/50 W (Operating)/< 10 W (Hot Standby)
    Power (INTL): + 15 to + 24 VDC/ 50 W (Operating)/ < 7 W (Hot Standby)
24. Operating Temp: - 40° C to + 75° C
25. Control I/O
    Gigabit Ethernet
26. Power I/O
    Snap Lock 12 Pin Connector
27. Data Output
    R/V Maps: 40 MBps
    Detections: 1 MBps
    Measurements: 1 MBps
    Tracks: 25 KBps
28. Mounting
    VESA 75 & 100mm